Web-Based Test Checker System for CCIT Department of North Valley College Foundation Inc

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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CHAPTER I
INTRODUCTION

Background of the Study

In today's educational system, technology plays a vital role in improving academic processes and reducing the limitations of manual work. Traditional methods of checking tests and quizzes are often time-consuming, prone to human error, and become inefficient when dealing with a large number of students. Because of these challenges, many institutions are now adopting digital and web-based solutions to enhance speed, accuracy, and reliability in academic evaluation.

At North Valley College Foundation Inc., specifically within the CCIT (College of Computing and Information Technology) Department, instructors manage multiple classes and assessments. As student enrollment continues to grow, manually checking examinations becomes increasingly difficult and may lead to delayed results, inconsistencies in grading, and difficulties in retrieving student exam records.

To address these concerns, this study proposes the development of a Web-based Test Checker System for the CCIT Department of North Valley College Foundation Inc. This system is designed to automate the checking and creating quizzes and examinations, provide faster result processing, and improve the accuracy of grading. It also aims to reduce the workload of instructors while ensuring efficient and fair evaluation of student performance.

Through this system, instructors will be able to manage answer keys, process student responses automatically, and generate test results with ease. In addition, it provides better record management and accessibility since it can be accessed through a web browser anytime and anywhere with an internet connection.

STATEMENT OF THE PROBLEM

The current process of test checking in the CCIT Department of North Valley College Foundation Inc. presents several challenges that affect efficiency, accuracy, and data management.

1. Inefficient and time-consuming manual checking of quizzes and examinations.
2. Prone to human errors in checking and computation of scores, leading to inaccurate grading and inconsistencies in test results.
3. Difficulty in organizing, storing, and retrieving student examination records.

Objectives of the Study

1. To design a web-based system that enables instructors to scan written quizzes and examination answer sheets for automated checking and scoring.
2. To develop a system that ensures accurate computation and reduces errors in grading results.
3. To create a database-driven system that securely stores, organizes, and retrieves examination records efficiently.

Scope and Delimitations of the Study

This study focuses on the development of a Web-based Test Checker System for the CCIT Department of North Valley College Foundation Inc. The system covers the scanning of written shaded quizzes and examination answer sheets for automated checking and scoring. It allows instructors to scan student answer sheets, process them through the system, and generate accurate results. The system also includes features for managing answer keys, storing student records, organizing class sections, and retrieving examination results through a centralized database. In addition, it

requires a stable internet connection to ensure proper access and smooth functionality of the web-based platform.

However, the system is limited only to scanned or shaded written quizzes and examinations and does not support fully online or browser-based exams where students answer directly in the system. It also does not include essay-type answer checking or advanced artificial intelligence-based evaluation. Furthermore, the system is designed specifically for the CCIT Department of North Valley College Foundation Inc. and may not fully support other departments or institutions without modification.

Significance of the Study

CCIT Teachers/Administrators	The system reduces the time and effort required in checking examinations and computing scores, improving efficiency and productivity.
CCIT Students	Students will receive faster results and immediate feedback, helping them evaluate and improve their academic performance.
North Valley College Foundation Inc	The institution will benefit from improved academic processes, better data management, and a more modernized assessment system.
Future Researchers	This study may serve as a reference for future researchers who plan to improve and enhance the system by adding more advanced features such as support for essay-type examinations, mobile accessibility, real-time analytics, artificial intelligence-based checking, and stronger security measures for managing academic records and examination results.

Chapter II
REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies that will serve as a guide in developing the system and data gathering.

According to Cuerdo et al. (2021), Argue that Philippine teachers are often overwhelmed with workload due to the high volume of clerical work, especially in evaluating student examinations, which may result in teacher burnout and turnover. The purpose of their study was to solve the problem through automation in test evaluation using a different Optical Mark Recognition (OMR) system known as EvalBee. Considering that schools have budget constraints for buying costly OMR systems, the researchers sought to determine if the cheaper alternative would be able to fulfill evaluation standards. Utilizing quantitative-evaluative research, they discovered that the use of EvalBee was effective, efficient, and reliable in evaluating examinations.

According to the study of Lainez, S. M. D. et al. (2024), An Optical Mark Recognition (OMR) is an effective data entry technology commonly used in educational and testing environments for capturing marked responses from documents such as surveys and examination sheets. The researchers developed an automated system to improve the speed and accuracy of checking entrance examinations for incoming university students, making the enrollment process more efficient. By replacing manual grading procedures, the system significantly accelerated the assessment of a 250-item multiple-choice examination.

According to the study of Kanjalkar, P. et al. (2003), This system was aimed at designing a cheap and efficient test checker system for MCQ tests based on the Optical Mark Recognition technique but without expensive specialist scanning machines. Instead of traditional OMR equipment, the system involved the use of phones for effective scanning of the test answer sheets and thus making automatic test checking available for small schools, tutorial centers, and individual lecturers. This system is designed to allow users to easily upload their files, analyze the results of students' tests

through image processing, and get exact scores, grades, and percentages. It not only provides feedback but also indicates whether the answer is correct, wrong, or actual using the color scheme. Besides, the system is equipped with a dashboard where analysis of student performance can be done while offering customizable OMR sheets depending on the required number of questions, answer options, and scoring. With respect to efficiency testing on 300 OMR sheets containing 12,800 questions, the system gave an average accuracy rate of 99.12%.

Based on Agarwal, S. et al. (2003), An automated testing checker system based on optical mark recognition is proposed, which is meant to score a multiple-choice examination using a standard scanning machine or any other mobile devices. To achieve this, image processing techniques will be used to analyze and grade marked questions on the answer sheet that has a blank circle. This testing checker will automatically compute the total score obtained by the student and also save the results in the database. Moreover, the system will instantly generate a list of all students who scored low marks with just one click. This testing checker was implemented to create an affordable and more efficient tool compared to manually checking the answer sheets or using the OMR tools. Using OMR technology along with image processing methods enables the tester checker to give high accuracy while scoring answer sheets in case there were some errors when the sheets were scanned. It is possible to test large sets of questionnaires in this tool within short periods of time.

As noted by Adiga, S., & Manjunatha, H. T. (2024), Modern advancements in image processing-based grading systems show that automated OMR systems are capable of attaining extremely high accuracy levels (above 95%), while at the same time minimizing the amount of time needed to process large numbers of exams. Such systems are built to scan answer sheets, recognize marked answers, and compare them against the provided key, thus making them very useful in academic settings.

According to Singh, J. et al., (2024), This research paper stresses the significance of having a system that would allow quick grading of multiple-choice questions through webcam. Despite the popularity of MCQ test systems owing to its convenience, manually grading answer sheets can be laborious, prone to errors, and challenging to handle in large quantities since the sheets might suffer from damage.

Though certain schools employ Optical Mark Recognition (OMR) machines, many other small schools find them costly to procure. Hence, the proposed solution in the study is the development of an economical system through the use of webcams and image processing methods, such as OpenCV, which could automatically assess answer sheets.

According to Jain, V. et al., (2022), This study target the improvement of a test checker system that applies optical mark recognition (OMR). The current grading systems have been expensive and needed specific scanners for the processing. Errors may occur when the sheets are not positioned correctly, the scanner is wrongly handled, or if there is damage to the answer sheets. These challenges can be solved by applying an image processing algorithm, which will enable automatic evaluation of the answer sheets without any errors. It will be reliable because it will address issues like angle distortion, blurring, and illumination. Besides, the improved system will be economical and portable because it can handle the processing of images collected by mobile phones instead of scanners.

CHAPTER III
METHODOLOGY

Technicality of the Project

The proposed Web-based Test Checker System for the CCIT Department of North Valley College Foundation Inc. will be developed using modern web technologies combined with Optical Mark Recognition (OMR) and image processing techniques to provide an efficient, accurate, and automated checking process for quizzes and examination answer sheets. The system will function as a web-based platform where instructors can create examinations, manage answer keys, scan student answer sheets, and automatically process results through OMR technology. It will be accessible through a web browser and will require a stable internet connection for proper functionality and secure data management.

The core feature of the system is the integration of Optical Mark Recognition (OMR), a technology used to detect shaded or marked answers on printed examination sheets. Through the use of image processing algorithms, the system will capture scanned images of student answer sheets using a scanner or mobile camera, identify alignment markers, detect answer bubbles, analyze marked responses, and compare them with the stored answer keys in the database. This process minimizes human error, improves checking speed, and ensures reliable scoring accuracy even when handling large numbers of examination papers.

The system will be developed using HTML, CSS, and JavaScript for the front-end interface. HTML will be used to structure the content of the system, CSS will handle the design and layout, and JavaScript will provide interactivity and dynamic features for a better user experience. For the back-end, PHP will be used as the server-side scripting language to handle system processes, manage data transactions, and control user authentication. MySQL will serve as the database management system for storing

student information, examination records, answer keys, and computed results in an organized and secure manner.

The system will work by allowing instructors to log in to their accounts and scan students' shaded answer sheets directly using a scanning device or camera integrated into the system. After scanning, the system will process the image and recognize the marked answers. It will then compare the responses with the stored answer key in the database and automatically compute the scores. The results will be generated instantly and stored in the system for record keeping, viewing, and reporting purposes. Instructors will also be able to manage answer keys, organize student records, and monitor class performance through the system interface.

To develop the system, various tools and technologies will be required. A local development environment such as XAMPP will be used to integrate Apache Server, PHP, and MySQL for system development and testing. A code editor like Visual Studio Code will be used for programming and system design. Web browsers such as Google Chrome or Microsoft Edge will be used for testing system functionality and compatibility. A scanning device or mobile camera will also be required for capturing the shaded answer sheets for processing within the system.

Through the integration of these technologies, tools, and technical expertise, the system aims to provide an efficient, reliable, and automated solution for checking scanned written quizzes and examinations in the CCIT Department.

Locale of the Study

In Figure 1, shows the location where the project was conducted in North Valley College Foundation Inc. located in Brgy Lanao Old Kidapawan in front of DPWH.

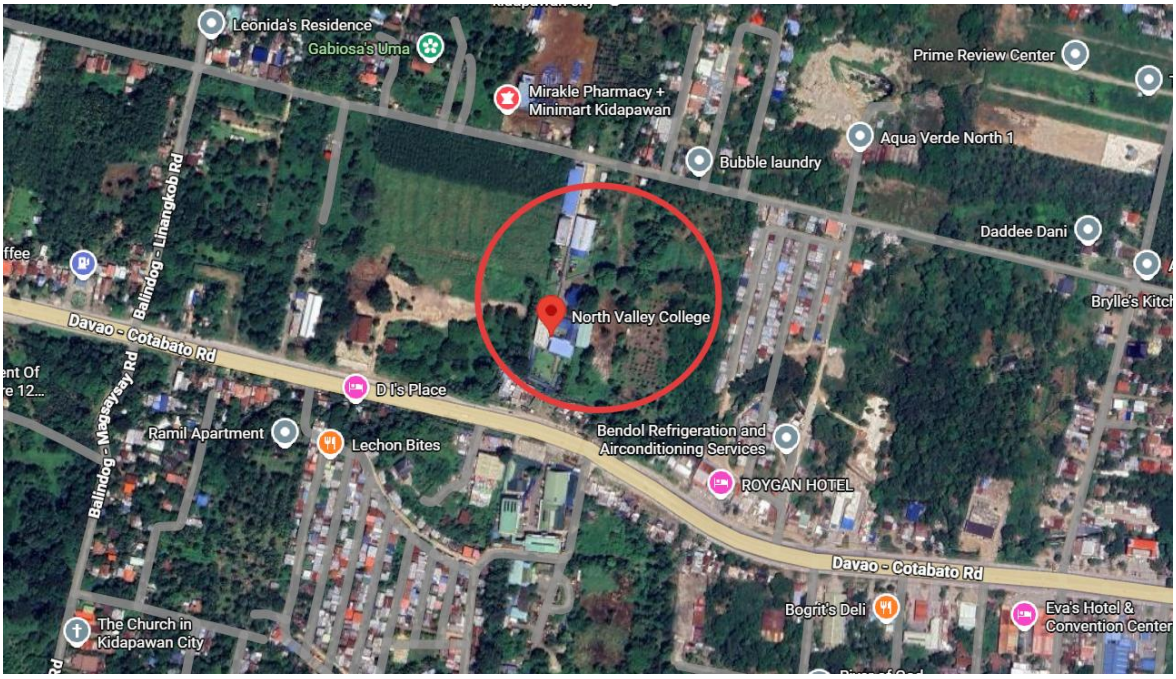


Figure 1: **North Valley College Foundation Inc**

Organizational Chart

At the highest level is the Program Head of the CCIT Department, who oversees the academic implementation and ensures that the system supports the department's evaluation processes and standards. Under the Program Head are the Instructors, who are the primary users of the system and are responsible for creating examinations, preparing answer keys, organizing class sections, and managing the scanning and checking of student answer sheets. The Students are the end-users who take the written quizzes and examinations using shaded answer sheets.

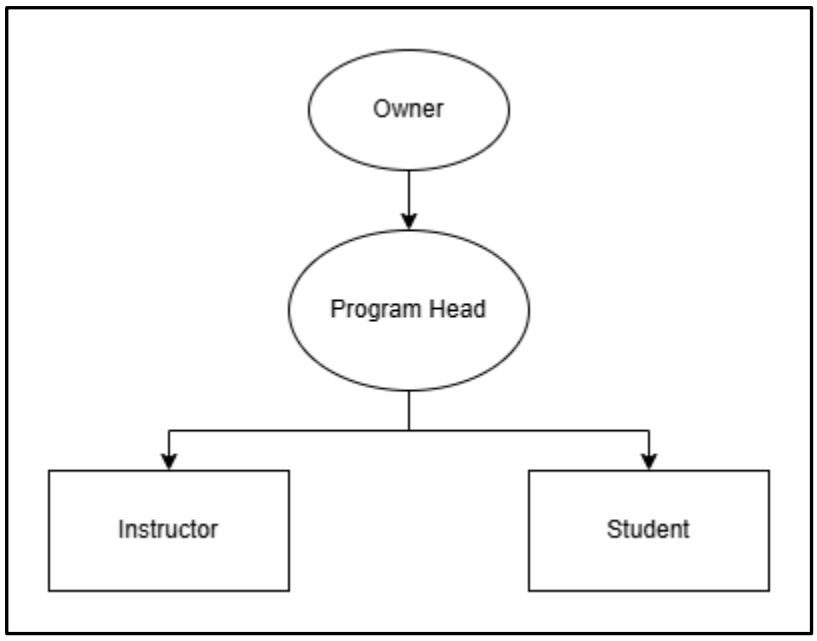


Figure 2: **Organizational Chart**

Requirements for Modeling

Figure 3 presents the Input-Process-Output (IPO) Chart of the Web-Based Test Checker System. The input stage includes examination details, answer keys, student information, class or section data, and scanned shaded answer sheets provided by instructors and students. During the process stage, the system manages examinations, scans answer sheets using OMR technology, processes student responses, compares answers with stored answer keys, computes scores, and stores records in the database. The output stage generates accurate examination results, score reports, and organized academic records for instructors and students. This model illustrates how the system transforms raw examination data into efficient, automated, and reliable grading results.

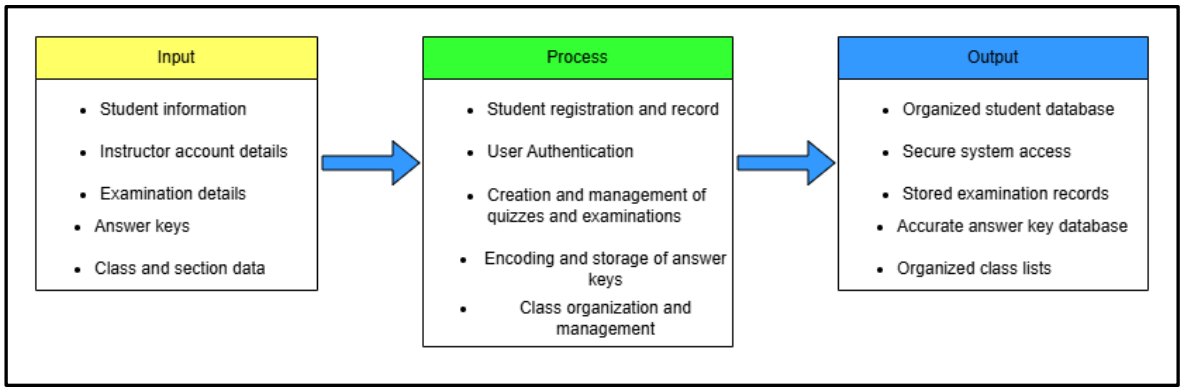


Figure 3: **IPO Chart**

Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) shows how the data in the system is organized and how each part is connected to one another. It helps explain how information flows inside the web-based test checker system.

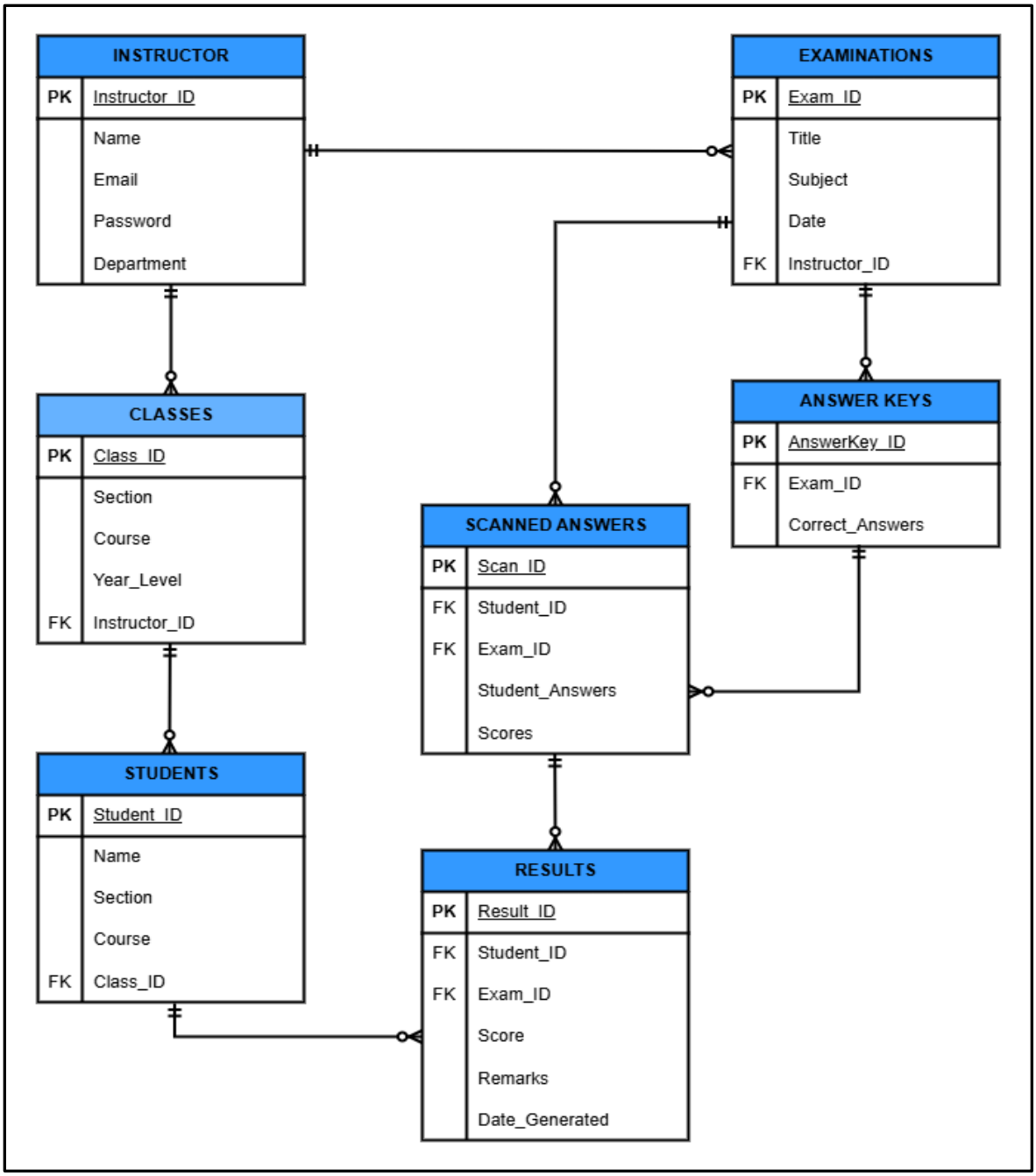


Figure 4: **Entity – Relationship Diagram**

Data Dictionary

The data dictionary defines the structure of the database used in the Web-based Test Checker System. It contains detailed information about each table, including field names, definitions, data types, sizes, constraints, nullability, and key assignments. Primary keys uniquely identify each record, while foreign keys establish relationships between tables. This organized structure ensures secure storage, efficient data processing, accurate record retrieval, and proper database management throughout the system.

The Instructors table contains the records of all instructors who use the system. It stores their personal information, login credentials, department assignment, and connection to the Program Head. This table is essential for managing examinations, answer keys, and student records.

Table 1

Instructor Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
instructor_id	INT	11	Unique identifier of the instructor	Auto Increment	Not Null	No	PK
name	VARCHAR	100	Full name of the instructor	None	Required	No	-
email	VARCHAR	100	Official email address	None	Unique	No	-
password	VARCHAR	255	Account password	None	Required	No	-
department	VARCHAR	100	Department assignment	CCIT	Required	No	-

The Students table stores student information, including names, sections, courses, and assigned instructors. It is used to organize students by class and maintain accurate academic records for examinations and performance monitoring.

Table 2

Student Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
student_id	INT	11	Unique student identifier	Auto Increment	Not Null	No	PK
name	VARCHAR	100	Full name of the student	None	Required	No	-
section	VARCHAR	50	Assigned class section	None	Required	No	-
course	VARCHAR	100	Student course/program	None	Required	No	-
instructor_id	INT	11	Assigned instructor ID	None	Foreign Key	No	FK

The Examinations table contains all quiz and exam details created by instructors, such as exam titles, subjects, schedules, and related instructor information. This table serves as the primary record for all assessments managed by the system.

Table 3

Examination Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
exam_id	INT	11	Unique examination identifier	Auto Increment	Not Null	No	PK
title	VARCHAR	100	Examination title	None	Required	No	-
subject	VARCHAR	100	Subject name	None	Required	No	-
exam_date	DATE	-	Date of examination	Current Date	Required	No	-
instructor_id	INT	11	Creator instructor ID	None	Foreign Key	No	FK

The Answer Keys table stores the correct answers for each examination. These answer keys are used by the system to automatically compare scanned student responses and calculate accurate scores.

Table 4

Answer Keys Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
answerkey_id	INT	11	Unique answer key identifier	Auto Increment	Not Null	No	PK
exam_id	INT	11	Linked examination ID	None	Foreign Key	No	FK
correct_answers	TEXT	Variable	Stored correct answers for checking	None	Required	No	-

The Scanned Answers table records the scanned student answer sheets, recognized responses, linked student and exam IDs, and initial computed scores. It functions as the main processing table for automatic test checking.

Table 5

Scanned Answer Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
scan_id	INT	11	Unique scanned record identifier	Auto Increment	Not Null	No	PK
student_id	INT	11	Linked student ID	None	Foreign Key	No	FK
exam_id	INT	11	Linked examination ID	None	Foreign Key	No	FK
student_answers	TEXT	Variable	Recognized student responses	None	Required	No	-
score	INT	11	Automatically computed score	0	Required	No	-

The Results table stores the final examination outcomes, including computed scores, remarks, and generated dates. It provides organized access to student performance reports and historical examination records for instructors and administrators.

Table 6

Result Table

Field Name	Data Type	Size	Description	Default	Constraints	Null	Key
result_id	INT	11	Unique result identifier	Auto	Not Null	No	PK
student_id	INT	11	Linked student ID	None	Foreign Key	No	FK
exam_id	INT	11	Linked examination ID	None	Foreign Key	No	FK
score	INT	11	Final computed score	0	Required	No	-
remarks	VARCHAR	50	Pass or fail remarks	None	Optional	Yes	-
date_generated	DATETIME	-	Date and time result was generated	Current Timestamp	Required	No	-

Data and Processing Modeling

Context Diagram

In this diagram, the central process is labeled ‘Web-Based Test Checker System,’ which represents the entire system responsible for scanning answer sheets, processing responses, computing scores, storing records, and generating examination results.

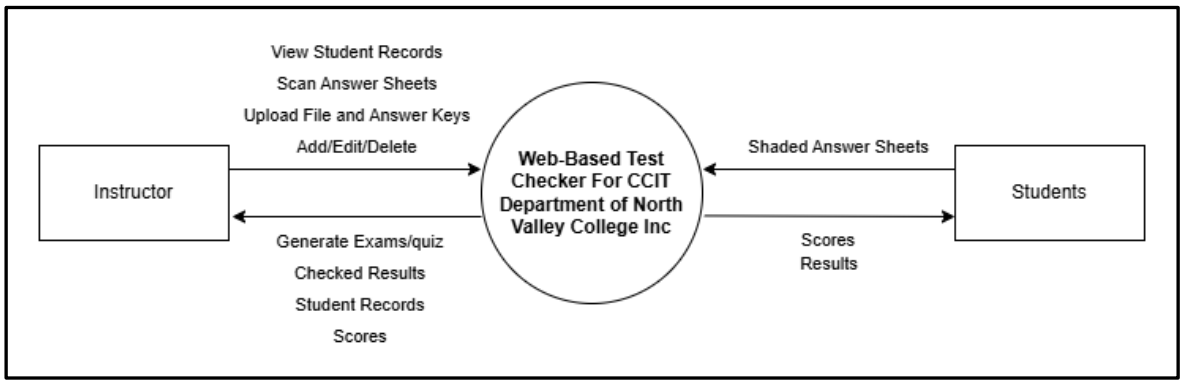


Figure 5: **Context Diagram**

Data Flow Diagram

The figure shown the Data Flow Diagram (DFD) of the Web-Based Test Checker System. The instructor inputs examination details, answer keys, and class information through the Manage Examination process, where data is stored in the system databases. Students submit shaded answer sheets, which are scanned and processed using OMR technology to capture responses accurately. The system then compares scanned answers with stored answer keys, computes scores automatically, and stores the results in the results database. Finally, the system generates and displays examination results and reports for both instructors and students. This process ensures faster, more accurate, and efficient test checking and record management.

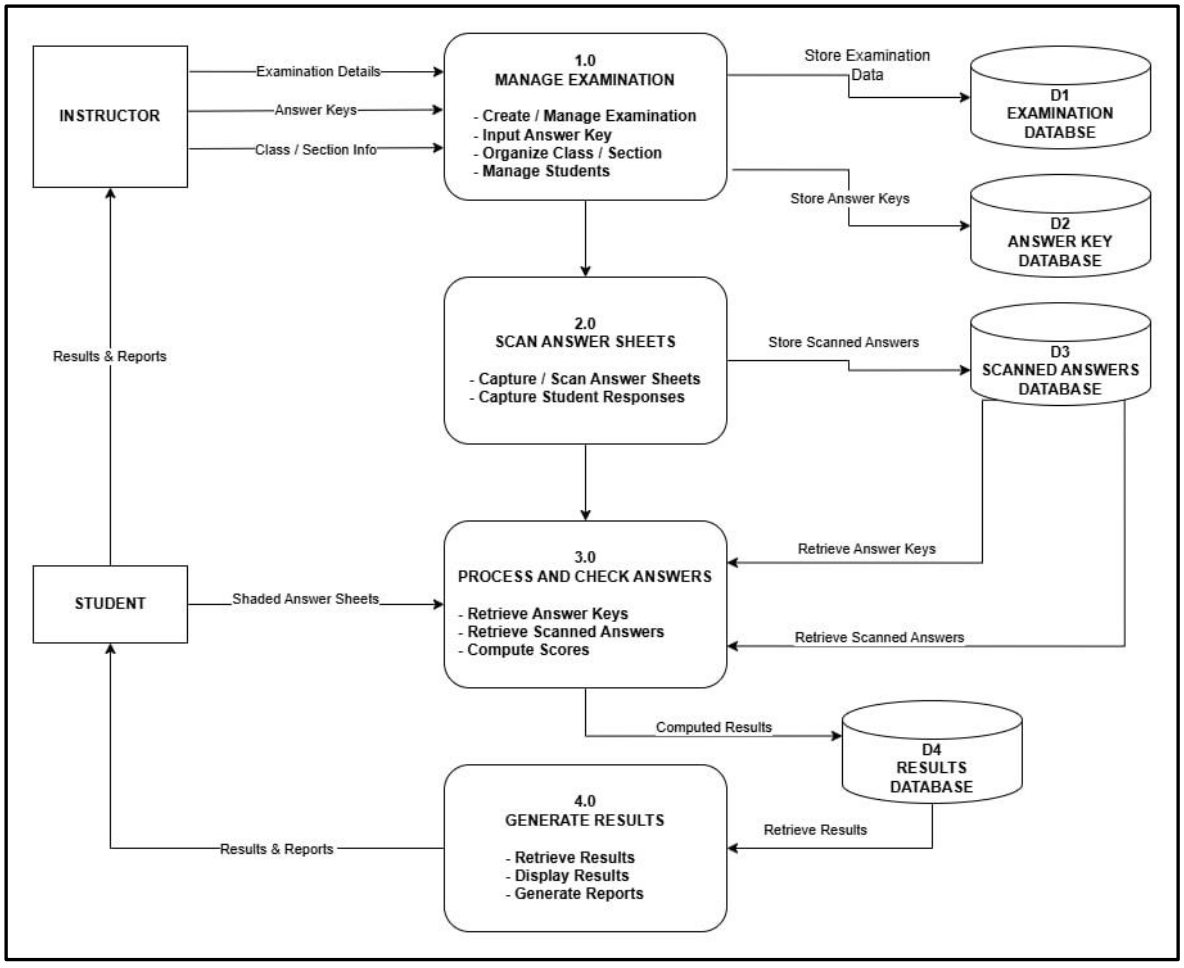


Figure 6: **Data Flow Diagram**

System Flowchart

The figure shown the flow of the Web-Based Test Checker System. The process starts with instructor login and account verification. Once validated, the instructor can manage exams and quizzes, student records, and class or section information. Instructors can create exams, edit answer keys, scan answer sheets, process results, and manage student data. After completing these tasks, the system stores the information and generates results efficiently. This flow ensures an organized and automated examination management process.

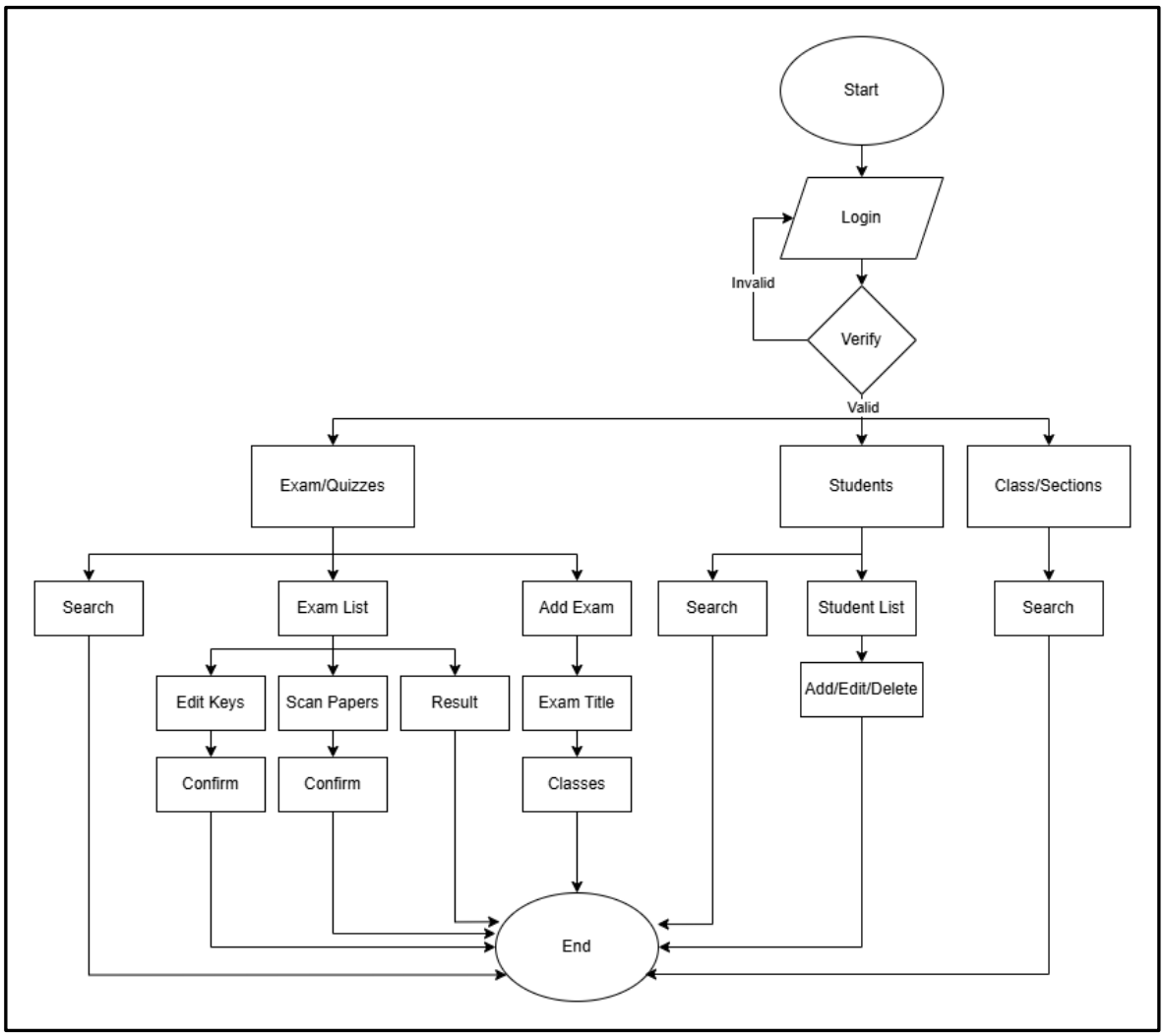


Figure 7: **SYSTEM FLOWCHART**

Disaster Recovery Plan

The Figure shown the Disaster Recovery Plan of the Web-Based Test Checker System. It outlines the system's goals of protecting examination data, ensuring accessibility, minimizing downtime, maintaining OMR functionality, and securing academic records. The plan identifies potential threats such as hacking, data loss, hardware failure, and system malfunctions, while providing recovery strategies including regular backups, security controls, backup devices, system maintenance, and quick restoration procedures to ensure continuous and reliable system operation.

Goals	Potential Threats	Response / Recovery Plan
• Protect examination data integrity	• Hacking, data loss, human error	• Regular backups, security systems, restricted access
• Ensure system accessibility	• Internet outage, server crash, power failure	• Backup internet, cloud hosting, UPS, manual backup process
• Minimize operational downtime	• Hardware failure, software malfunction	• System maintenance, spare devices, quick restoration
• Maintain OMR scanning and automated grading system	• Scanner failure, image processing errors, corrupted scanned data	• Backup scanning devices, regular testing, manual checking option
• Secure examination results and academic records	• Data corruption, unauthorized access	• Automated backups, admin-only access, audit logs

Figure 8: **Disaster Recovery Plan**

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